THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHEMATICS TWO-CHAPTER TEST SET D

Chapters 5 & 7: **Arithmetic Progressions, Coordinate Geometry** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The first four terms of the AP with $a = 10$, $d = 10$ are: 1

(a) $0, 10, 20, 30$ **(b)** $10, 20, 30, 40$ **(c)** $10, 11, 12, 13$ **(d)** $10, 100, 1000, \ldots$

2. The 10th term of the AP $\frac{1}{2}, 1, \frac{3}{2}, \ldots$ is: 1

(a) 5 **(b)** $\frac{11}{2}$ **(c)** $\frac{9}{2}$ **(d)** 10

3. The 6th term of the AP with $a_n = 4n - 3$ is: 1

(a) 25 **(b)** 21 **(c)** 24 **(d)** 17

4. A student states: "$1 + 2 + \cdots + n = n^2$." The correct response is: 1

(a) Correct **(b)** Incorrect — it is $\frac{n(n+1)}{2}$

(c) Correct for odd n **(d)** Cannot be determined

5. For the number of multiples of 4 between 10 and 250, Student A says 60; Student B says 61. Then: 1

(a) Student B is correct **(b)** Student A is correct

(c) Both are correct **(d)** Both are wrong

6. The distance between $(-1, -1)$ and $(2, 3)$ is: 1

(a) 5 **(b)** 7 **(c)** $\sqrt{7}$ **(d)** 25

7. The mid-point of the join of $(-4, 5)$ and $(6, -1)$ is: 1

(a) $(1, 2)$ **(b)** $(5, 3)$ **(c)** $(-1, 2)$ **(d)** $(1, 3)$

8. A point on the x-axis at distance 5 from $(2, 3)$ is: 1

(a) $(3, 0)$ **(b)** $(6, 0)$ **(c)** $(5, 0)$ **(d)** $(7, 0)$

9. A student states: "Every point on the y-axis has abscissa 1." The correct response is: 1

(a) Incorrect — the abscissa is 0 **(b)** Correct

(c) Its ordinate is 0 **(d)** Cannot be determined

10. If P(4, m) is the mid-point of A(2, 3) and B(6, −3), Student A says $m = 0$; Student B says $m = 3$. Then: 1

(a) Student B is correct **(b)** Student A is correct

(c) Both are correct **(d)** Both are wrong

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the sum $1 + 3 + 5 + \cdots + 199$. 2

12. The 5th and 9th terms of an AP are 13 and 25. Find a and d. 2

13. Show that $(1, -1)$, $(-1, 1)$ and $(-\sqrt{3}, -\sqrt{3})$ are the vertices of an equilateral triangle. 2

14. If M(5, −1) is the mid-point of AB and A is (2, 3), find B. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. The angles of a triangle are in AP and the smallest angle is $40°$. Find the other angles. 3

16. In what ratio does P(−4, 6) divide the join of A(−6, 10) and B(3, −8)? 3

17. An AP has 37 terms. The sum of the three middle-most terms is 225 and the sum of the last three terms is 429. Find the AP. 3

OR

Find the sum of all two-digit odd numbers. 3

18. If A(−2, −1), B(a, 0), C(4, b) and D(1, 2) are the vertices of a parallelogram, find a and b. 3

OR

The segment joining A(2, 1) and B(5, −8) is trisected at P and Q, P being nearer A. If P lies on the line $2x - y + k = 0$, find k. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study:** Cans are stacked in a supermarket with 30 cans in the bottom row, 28 in the next, and so on, with 2 cans in the top row. 5

(i) Find the number of rows.

(ii) Find the total number of cans.

(iii) How many cans are in the 10th row from the bottom?

OR

The sum of the 3rd and 7th terms of an AP is 6 and their product is 8. Find the sum of the first 16 terms. 5

20. **Case study:** In a village map, a well is at W(2, 3) and a temple at T(8, 11). 5

(i) Find WT.

(ii) A school lies on WT dividing it in the ratio 3 : 2 from W. Find its position.

(iii) A tap on the x-axis is equidistant from W and T. Find it.

OR

Show that A(−1, 0), B(3, 1), C(2, 2) and D(−2, 1) are the vertices of a parallelogram. Is it a rectangle? Justify. 5

END OF QUESTION PAPER — Total: 40 Marks